

Artificial Neural Networks and Deep Learning

Assignment 3

Guidelines

- Show all steps clearly by hand.
- Write formulas before calculations.
- Draw diagrams where needed.
- Keep answers neat and structured.

Part 1: Dense Prediction with CNNs

Q1. (Conceptual – What is Dense Prediction?)

Define **dense prediction**.

How is it fundamentally different from:

- Image classification
- Object detection

Give one real-world example where dense prediction is essential.

Q2. (Conceptual – CNN Limitation for Dense Tasks)

Standard CNNs reduce spatial resolution due to pooling.

- a) Why does this happen?
- b) Why is this a problem for segmentation tasks?
- c) What kind of information is lost?

Q3. (Numerical – Downsampling Effect)

An image of size 256×256 passes through 3 pooling layers (2×2 , stride 2).

- Compute final feature map size
- What percentage of spatial resolution is retained?
- Explain impact on pixel-level prediction

Q4. (Conceptual – Fully Convolutional Networks (FCN))

Explain how a classification CNN is converted into an FCN.

- What happens to fully connected layers?
- Why does this allow variable input size?

Q5. (Numerical – Converting FC to Conv)

A fully connected layer takes input of size $7 \times 7 \times 512$ and outputs 4096 neurons.

- Convert this into an equivalent convolutional layer
- What is the filter size and number of filters?
- What is the output dimension?

Q6. (Conceptual – Upsampling Methods)

Explain different ways to increase spatial resolution:

- Nearest neighbor upsampling
- Bilinear interpolation
- Transposed convolution

Compare them in terms of:

- Learnability
- Accuracy
- Complexity

Q7. (Numerical – Transposed Convolution)

A feature map of size 16×16 is passed through a transposed convolution layer with:

- Stride = 2
- Kernel size = 4

- a) Estimate output size (conceptually)
- b) Why does this operation increase resolution?

Q8. (Conceptual – Skip Connections / U-Net Idea)

Explain why **skip connections** are used in dense prediction models.

- What is combined?
- Why are early layer features important?

Relate your answer to **localization vs semantic understanding**.

Q9. (Conceptual + Numerical – Output Representation)

For semantic segmentation with:

- Image size = 128×128
- Number of classes = 4

- a) What is the output tensor size?
- b) What does each pixel output represent?
- c) Which activation function is used and why?

Q10. (Design Question – Full Pipeline)

Design a CNN architecture for **semantic segmentation** of road scenes. Your answer should include:

- Input layer
- Downsampling (encoder)
- Bottleneck
- Upsampling (decoder)
- Skip connections
- Final output layer

Explain flow step-by-step in simple terms.

Part 2: Recurrent Neural Networks (RNNs)

Q11. (Conceptual – Why RNNs?)

Why are traditional feedforward neural networks not suitable for sequence data?

Explain how RNNs solve this problem.

Give two real-world examples (e.g., text, speech, time series).

Q12. (Conceptual – RNN Architecture)

Draw a basic RNN unit and explain:

- Input x_t
- Hidden state h_t
- Output o_t

What is the role of the hidden state in remembering past information?

Q13. (Numerical – RNN State Calculation)

Given:

- $h_t = \tanh(W_h h_{t-1} + W_x x_t)$
- $h_{t-1} = 0.5, x_t = 1$
- $W_h = 0.8, W_x = 0.2$

- a) Compute the input to activation
- b) Compute h_t (use tanh approximation if needed)
- c) Explain what this value represents

Q14. (Conceptual – Unrolling RNNs)

Explain what it means to **unroll an RNN through time**.

- Why is unrolling important for training?
- How does it relate to backpropagation?

Q15. (Conceptual – Types of RNN Tasks)

Explain the following RNN input-output structures:

- One-to-one
- One-to-many
- Many-to-one
- Many-to-many

Give one example for each.

Q16. (Conceptual – Backpropagation Through Time (BPTT))

What is **Backpropagation Through Time (BPTT)**?

- How is it different from standard backpropagation?
- Why is it computationally expensive?

Q17. (Conceptual – Vanishing Gradient Problem)

Explain the **vanishing gradient problem** in RNNs.

- Why does it occur?
- What is its effect on learning long-term dependencies?

Q18. (Conceptual – Exploding Gradient Problem)

What is the **exploding gradient problem**?

- How does it affect training?
- Suggest one method to handle it

Q19. (Conceptual – LSTM Motivation)

Why were **LSTMs (Long Short-Term Memory networks)** introduced?

- What problem do they solve?
- Briefly explain idea of memory cell and gates

Q20. (Design + Understanding – Sequence Modeling)

Design an RNN-based model for **sentiment analysis of a sentence**.

Your answer should include:

- Input representation (words → vectors)
- RNN processing sequence
- Final output layer
- Type of RNN structure used

Explain flow step-by-step.